

Water Dam Monitoring with Weather Integration

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Abstract. The abstract should briefly summarize the contents of the paper in 150–250 words.

Keywords: First keyword · Second keyword · Another keyword.

1 Introduction

The management of hydroelectric dams represents a critical engineering challenge, as these infrastructures are vital for both sustainable water supply and electricity generation. This project is motivated by the inherent risks associated with dam operations: the danger of overflow during periods of heavy rainfall, which presents significant flooding and safety risks, and the threat of drought during dry seasons, which leads to reduced energy production. Currently, monitoring methods can often be inefficient or outdated. Consequently, there is a clear need for a smart, autonomous, and remote monitoring system to unify and optimize these industrial processes.

The primary objective of this project is to develop a Cyber-Physical System (CPS) and Internet of Things (IoT) framework for dam monitoring and control. The proposed solution collects real-time data on variables such as water level, flow rates and rainfall intensity, supporting compliance with the legally required safety control and observation measures defined for each dam class and associated damage potential, in accordance with the Portuguese dam safety framework established by Decreto-Lei n.^o 21/2018, de 28 de maro[1], which regulates the safety of large and small dams throughout the phases of design, construction, first impoundment and operation. While the legislation does not prescribe a universal operating range in terms of reservoir percentage, this work adopts a conservative operational strategy in which the CPS helps to keep the reservoir within project-defined safe limits (for example, between 20% and 90% of storage capacity). This is achieved through a distributed architecture utilizing Arduino Nano ESP32 units for local data acquisition and a Raspberry Pi gateway running a Node-RED server and MQTT broker. Furthermore, the system supports predictive decisions using weather forecasts and historical data processed via Gradient Boosting Machine (GBM) models, all visualized through a remote Node-RED dashboard for manual and automatic gate control.

2 Fundamental Knowledge and Related Work

2.1 Review of Existing Systems

2.2 Gap Analysis

3 Proposed Solution

3.1 System Design and Architecture

3.2 Implementation Details

4 Evaluation and Results

4.1 Testing Methodology

4.2 Case Studies

4.3 Result Analysis

5 Discussion

5.1 Benefits

5.2 Limitations

6 Conclusion and Future Work

Acknowledgments. A bold run-in heading in small font size at the end of the paper is used for general acknowledgments, for example: This study was funded by X (grant number Y).

Disclosure of Interests. It is now necessary to declare any competing interests or to specifically state that the authors have no competing interests. Please place the statement with a bold run-in heading in small font size beneath the (optional) acknowledgments¹, for example: The authors have no competing interests to declare that are relevant to the content of this article. Or: Author A has received research grants from Company W. Author B has received a speaker honorarium from Company X and owns stock in Company Y. Author C is a member of committee Z.

References

1. República Portuguesa: Decreto-Lei n.º 21/2018, de 28 de março — Aprova o regime de segurança de barragens e o Regulamento de Pequenas Barragens. In: *Diário da República*, 1.ª série, n.º 62, pp. 1368–1394. Imprensa Nacional-Casa da Moeda, Lisboa (2018)

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